Alternative formulation (similar to Chap. 6 of Melrose–McPhedran)

Limitations of separating induced current into $\frac{\partial \vec{P}}{\partial t}$ and $\vec{\nabla}\times\vec{M}$:

$$\vec{j}_{ind} = \frac{\partial \vec{P}}{\partial t} + \vec{\nabla}\times\vec{M}$$

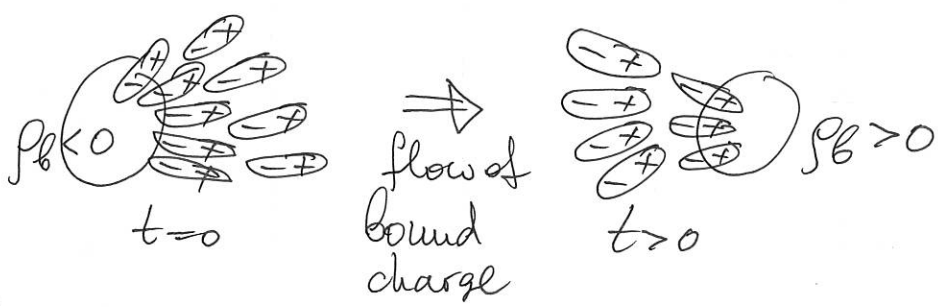
Observe that $\vec{\nabla}\cdot\vec{j}_{ind} = \frac{\partial}{\partial t}\vec{\nabla}\cdot\vec{P} + \underbrace{\vec{\nabla}\cdot(\vec{\nabla}\times\vec{M})}_{\phi} = -\frac{\partial}{\partial t}\rho_b$

$$\left(\rho_b = -\vec{\nabla}\cdot\vec{P}\right)$$

and therefore $\vec{\nabla}\times\vec{M}$ does not contribute to the change of the bound charge density distribution.

In other words, $\vec{\nabla}\times\vec{M}$ describes closed current loops

while $\frac{\partial \vec{P}}{\partial t}$ describes sources and sinks of bound charges



$$\begin{cases} \vec{\nabla}\cdot\vec{B} = 0 \\ \vec{\nabla}\times\vec{E} + \frac{\partial \vec{B}}{\partial t} = 0 \\ \vec{\nabla}\cdot\vec{D} = \rho_f \\ \vec{\nabla}\times\vec{H} - \frac{\partial \vec{D}}{\partial t} = \vec{j}_f \end{cases}$$

Observe that $\vec{\nabla}\cdot\left(\vec{D} + \vec{\nabla}\times\vec{Q}\right) = \vec{\nabla}\cdot\vec{D}$

while

$$\vec{\nabla}\times\left(\vec{H} + \frac{\partial \vec{Q}}{\partial t}\right) - \frac{\partial}{\partial t}\left(\vec{D} + \vec{\nabla}\times\vec{Q}\right) =$$

$$= \vec{\nabla}\times\vec{H} - \frac{\partial \vec{D}}{\partial t} + \underbrace{\left[\frac{\partial}{\partial t}\vec{\nabla}\times\vec{Q} - \frac{\partial}{\partial t}\vec{\nabla}\times\vec{Q}\right]}_{=0}$$

Therefore definitions of $\vec{D}$ and $\vec{H}$ are not unique:

$\vec{D}' = \vec{D} + \vec{\nabla}\times\vec{Q}$ and $\vec{H}' = \vec{H} + \frac{\partial \vec{Q}}{\partial t}$ describe the same physics for arbitrary function $\vec{Q}(\vec{r},t)$.

More specifically, $\frac{\partial \vec{P}}{\partial t}$ describes <u>longitudinal</u> current $\vec{j}^{\parallel}_{ind}$:
(along vector $\hat{k} = \frac{\vec{k}}{k}$)

$$\vec{j}_{ind} = \vec{j}^{\parallel}_{ind} + \vec{j}^{\perp}_{ind}$$

where $\hat{k}\cdot\vec{j}^{\perp}_{ind} = 0$ while $\vec{j}^{\parallel}_{ind}\cdot\hat{k} = j^{\parallel}_{ind}$, $\hat{k} = \frac{\vec{k}}{k}$.

$$\Rightarrow \vec{\nabla}\cdot\vec{j}_{ind}(\vec{r},t) \overset{F.T.}{\Longrightarrow} \vec{k}\cdot\vec{j}_{ind}(\vec{k},\omega) = k\, j^{\parallel}_{ind}(\vec{k},\omega)$$

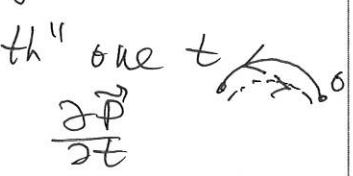
At the same time, $\vec{\nabla}\times\vec{M} \overset{F.T.}{\Longrightarrow} \vec{k}\times\vec{M}(\vec{k},\omega)$ describes <u>transverse</u> (to $\hat{k}$) current. $\vec{j}^{\perp}_{ind}$.

When $\vec{P}$ and $\vec{M}$ oscillate as $e^{i\omega t}$ with time we can distinguish $\vec{j}^{\parallel}_{ind} = \frac{\partial \vec{P}}{\partial t}$ and $\vec{j}^{\perp}_{ind} = \vec{\nabla}\times\vec{M}$ only on times longer than period $T = \frac{2\pi}{\omega}$:

during the period $\vec{\nabla}\times\vec{M}$ completes "rotation"

while $\frac{\partial \vec{P}}{\partial t}$ $\xrightarrow{t=0 \;\to\; t=T/2}$ $\xleftarrow{t=T}$ "traces" back to $t=0$ configuration.

But for $t < T$ we can not distinguish "arc of the circle" trajectory from the "back-and-forth" one



$\vec{P} = \varepsilon_0 \chi_e \vec{E}$, $\vec{D} = \varepsilon_0 (1+\chi_e) \vec{E} = \varepsilon \vec{E}$, $\vec{D} = \varepsilon_0 \vec{E} + \vec{P}$

$\vec{M} = \chi_m \vec{H}$

Induced current $\vec{j}_b = \frac{\partial \vec{P}}{\partial t} + \vec{\nabla} \times \vec{M}$

$\boxed{\text{Alternative formulation}}$: <u>generalized displacement</u> $\vec{\mathcal{D}}$

let $\vec{\mathcal{D}}_{(\vec{r},t)} = \varepsilon_0 \vec{E}_{(\vec{r},t)} + \int_{-\infty}^{t} dt' \, \vec{j}_b(\vec{r},t') = \varepsilon_0 \vec{E} + \vec{\mathcal{P}}(\vec{r},t)$

where <u>generalized polarization</u> $\vec{\mathcal{P}} = \int_{-\infty}^{t} dt' \, \vec{j}_b(r,t')$

Then $\vec{j}_b(r,t) = \frac{\partial \vec{\mathcal{P}}}{\partial t}$ and $\rho_b(r,t) = -\vec{\nabla} \cdot \vec{\mathcal{P}}$.

since $\vec{\nabla} \cdot \vec{j} + \frac{\partial \rho}{\partial t} = 0 \Rightarrow \frac{\partial}{\partial t}\left(\vec{\nabla} \cdot \vec{\mathcal{P}} + \rho\right) = 0$

Then Maxwell's equs read:

$$\vec{\nabla} \times \vec{E} + \frac{\partial}{\partial t} \vec{B} = 0 \qquad (1)$$

$$\vec{\nabla} \cdot \vec{B} = 0 \qquad (2)$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t} + \mu_0 \underbrace{\vec{J}}_{\vec{J}_{ext} + \vec{J}_b}$$

$$= \mu_0 \vec{J}_{ext} + \mu_0 \left(\varepsilon_0 \frac{\partial \vec{E}}{\partial t} + \frac{\partial \vec{\mathcal{P}}}{\partial t} \right)$$

$$\Rightarrow \vec{\nabla} \times \vec{B} = \mu_0 \vec{j}_{ext} + \mu_0 \frac{\partial \vec{\mathcal{D}}}{\partial t} \qquad (3)$$

$$\vec{\nabla} \cdot \vec{E} = \frac{1}{\varepsilon_0} \rho \Rightarrow \vec{\nabla} \cdot \left(\varepsilon_0 \vec{E}\right) = \rho_{ext} + \left(-\vec{\nabla} \cdot \vec{\mathcal{P}}\right)$$

$$\Rightarrow \vec{\nabla} \cdot \vec{\mathcal{D}} = \rho_{ext} \qquad (4)$$

And so we have 4 equs and 3 fields: $\vec{B}$, $\vec{E}$ and $\vec{\mathcal{D}}$.

Need to add constitutive relations which we write as

$$\vec{\mathcal{D}} = \tilde{\varepsilon}\, \vec{E}$$

(Note that in general $\varepsilon \neq \tilde{\varepsilon}$ but they are closely related.) Also note that this formulation does not require field $\vec{H}$! Everything is contained in $\vec{\mathcal{D}}$.

See discussion in Ch. 6, pp. 64-74, for more arguments. [Melrose, McPhedran book]

They write $\boxed{\text{eq. 6.15}}$ $(J_{ind})_i(\omega, \vec{k}) = \sigma_{ij}(\omega, \vec{k})\, E_j(\omega, \vec{k})$

where $\vec{J}_{ind}$ is the induced current $= \vec{J}_b$ in our notations.

From $\vec{\mathcal{D}} = \tilde{\varepsilon}\, \vec{E} = \varepsilon_0 \vec{E} + \vec{\mathcal{P}}$ we get

$$\vec{\mathcal{P}} = (\tilde{\varepsilon} - \varepsilon_0)\, \vec{E} \quad \text{and} \quad \vec{j}_b = \vec{j}_{ind} = \frac{\partial}{\partial t} \vec{\mathcal{P}}$$

$$\Rightarrow \vec{j}_{ind}(r,t) = \frac{\partial}{\partial t}(\tilde{\varepsilon} - \tilde{\varepsilon}_0)\, \vec{E}(r,t)$$

↳ this in general implies non-local in space-time relation.

Upon Fourier transform we get

$$(\vec{j}_{ind})_i(\omega, \vec{k}) = -i\omega(\tilde{\varepsilon}_{ij} - \varepsilon_0 \delta_{ij})\, E_j(\omega, \vec{k})$$

$$\Rightarrow \sigma_{ij}(\omega, k) = -i\omega(\tilde{\varepsilon}_{ij} - \varepsilon_0 \delta_{ij})$$

$$\tilde{\varepsilon}_{ij}(\omega,\vec{k}) = \varepsilon_0\,\delta_{ij} + \frac{\sigma_{ij}(\omega,\vec{k})}{-i\omega} = \varepsilon_0\,\delta_{ij} + \frac{i\,\sigma_{ij}(\omega,\vec{k})}{\omega}$$

$$= \varepsilon_0\left(\delta_{ij} + \frac{i\,\sigma_{ij}(\omega,\vec{k})}{\varepsilon_0\,\omega}\right) = \varepsilon_0\,K_{ij}(\omega,\vec{k})$$

$\boxed{\text{eq. 6.17}}$

<u>Melrose – McPhedran: Chap. 6</u>

$$K_{ij}(\omega,\vec{k}) = \delta_{ij} + \frac{i}{\varepsilon_0\,\omega}\,\sigma_{ij}(\omega,\vec{k}) = \delta_{ij} + \frac{1}{\varepsilon_0\,\omega^2}\,\alpha_{ij}(\omega,\vec{k})$$

$$= \delta_{ij} + \chi_{ij}(\omega,\vec{k})$$

$$\tilde{\varepsilon}_{ij}(\omega,\vec{k}) = \varepsilon_0\,K_{ij}(\omega,\vec{k})$$

equivalent dielectric tensor

equivalent permittivity tensor

The only difference with the standard (and historic) approach is that $\vec{\mathcal{D}} = \tilde{\varepsilon}\,\vec{E}$ follows from setting $\vec{M} = 0$ so that $\vec{\mathcal{P}} = \int^{t} dt'\,\vec{j}_{ind}$.

Note: this is all for AC fields! ~~[illegible]~~
Static fields are excluded!

$$D_\alpha(\vec{r},t) = \int_{-\infty}^{\infty} dt' \int d\vec{r}\,' \, \varepsilon_{\alpha\beta}(\vec{r}-\vec{r}\,', t-t') \, E_\beta(\vec{r}\,', t')$$

where $\varepsilon_{\alpha\beta}(\vec{\rho}, t-t') = 0$ for $t' > t$ (causality)

Then for $D_\alpha(\omega, \vec{k}) = \int_{-\infty}^{\infty} dt \int d\vec{r} \; e^{i\omega t - i\vec{k}\cdot\vec{r}} \, D_\alpha(\vec{r}, t)$

we obtain simple relation:

$$D_\alpha(\omega, \vec{k}) = \varepsilon_{\alpha\beta}(\omega, \vec{k}) \, E_\beta(\omega, \vec{k})$$

Note that the frequency is understood to have infinitesimaly small imaginary part: $\omega \to \omega + i\delta$ (causality)

The same holds for the generalized displacement

$$\mathcal{D}_\alpha(\omega, \vec{k}) = \tilde{\varepsilon}_{\alpha\beta}(\omega, \vec{k}) \, E_\beta(\omega, \vec{k})$$

Behavior of $\varepsilon(\omega)$ and $\tilde{\varepsilon}(\omega)$ (suppressing $\vec{k}$ dependence for now)

$$\cancel{\vec{D}(\omega) = \varepsilon_0 \vec{E}(\omega) + \int_{-\infty}^{t} dt' \, \vec{j}_{ind}(\vec{r}, t')}$$

$$\vec{D}(t) = \varepsilon_0 \vec{E}(t) + \int_{-\infty}^{t} dt' \, \vec{j}_{ind}(\vec{r}, t')$$

$$\int_{-\infty}^{\infty} dt \, e^{i\omega t} \vec{D}(t) = \vec{D}(\omega) = \varepsilon_0 \vec{E}(\omega) +$$

$$+ \int_{-\infty}^{\infty} dt \, e^{i\omega t} \int_{-\infty}^{t} dt' \underbrace{\vec{j}_{ind}(t')}_{\sigma \vec{E}(t')}$$

<u>Ohm's law</u>

σ = conductivity

But $\vec{E}(t') = \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} e^{-i\omega' t'} \vec{E}(\omega')$

$$\Rightarrow \int_{-\infty}^{\infty} dt \, e^{i\omega t} \int_{-\infty}^{t} dt' \int \frac{d\omega'}{2\pi} e^{i\omega t} e^{-i\omega' t'} \sigma \vec{E}(\omega')$$

$$= \int \frac{d\omega'}{2\pi} \int_{-\infty}^{\infty} dt \, e^{i\omega t} \underbrace{\left(\frac{e^{-i\omega' t'}}{-i\omega'} \Big|_{-\infty}^{t} \right)}_{\frac{e^{-i\omega' t} - e^{-i(\omega' + i\delta)T}}{-i\omega'} \Big|_{T \to -\infty} = \frac{e^{-i\omega' t}}{-i\omega}} \sigma \vec{E}(\omega')$$

$$= \int \frac{d\omega'}{2\pi} \int_{-\infty}^{\infty} dt \, \frac{e^{i(\omega - \omega')t}}{-i\omega'} \sigma \vec{E}(\omega') =$$

$$= \int d\omega' \, \delta(\omega - \omega') \frac{\sigma \vec{E}(\omega')}{-i\omega'} = \frac{i\sigma}{\omega} \vec{E}(\omega)$$

Generally $\vec{j}_{ind}(t') = \int_{-\infty}^{t'} dt''\, \sigma(t'-t'')\, \vec{E}(t')$ Ohm's law for frequency dependent conductivity.

$$\Rightarrow \vec{j}_{ind}(\omega) = \sigma(\omega)\, \vec{E}(\omega)$$

and $\vec{j}_{ind}(t') = \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} e^{-i\omega' t'} \vec{j}_{ind}(\omega')$

So $\vec{D}(\omega) = \varepsilon_0 \vec{E}(\omega) + \int_{-\infty}^{\infty} dt\, e^{i\omega t} \int_{-\infty}^{t} dt' \underbrace{\int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} e^{-i\omega' t'} \vec{j}_{ind}(\omega')}_{\vec{j}_{ind}(t')}$

$$= \varepsilon_0 \vec{E}(\omega) + \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} \int_{-\infty}^{\infty} dt\, e^{i\omega t}\, \vec{j}_{ind}(\omega') \underbrace{\int_{-\infty}^{t} dt'\, e^{-i\omega' t'}}_{\left.\frac{e^{-i\omega' t'}}{-i\omega'}\right|_{-\infty}^{t}}$$

Here $\lim_{T\to-\infty} e^{-i\omega' t'}\Big|_T^t = e^{-i\omega' t} - e^{-i\omega' T}\Big|_{T\to-\infty} =$

$= e^{-i\omega' t}$ since $\omega' = \omega' + i\delta$ $(\delta = +0)$

so that $e^{-i(\omega'+i\delta)T} = e^{-i\omega' T} e^{\delta T} \xrightarrow[T\to-\infty]{} 0$

$$\Rightarrow \vec{D}(\omega) = \varepsilon_0 \vec{E}(\omega) + \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} \int_{-\infty}^{\infty} dt\, e^{i\omega t} \frac{e^{-i\omega' t}}{-i\omega'} \vec{j}_{ind}(\omega')$$

$$= \varepsilon_0 \vec{E}(\omega) + \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi}\, 2\pi\, \delta(\omega-\omega') \frac{\vec{j}_{ind}(\omega')}{-i\omega'} =$$

$$= \varepsilon_0 \vec{E}(\omega) + \frac{i\, \vec{j}_{ind}(\omega)}{\omega} \quad \text{but } \vec{j}_{ind}(\omega) = \sigma(\omega)\vec{E}(\omega)$$

$$\vec{D}(\omega) = \varepsilon_0 \vec{E}(\omega) + \frac{i}{\omega}\sigma(\omega)\vec{E}(\omega) = \varepsilon_0\left(1 + \frac{i\sigma(\omega)}{\omega\varepsilon_0}\right)\vec{E}(\omega)$$

Thus $\vec{\mathcal{D}}(\omega) = \varepsilon_0 \vec{E}(\omega) + \frac{i\sigma}{\omega}\vec{E}(\omega) = \left(\sigma \to \sigma(\omega) \text{ in general}\right)$

$$= \left(\varepsilon_0 + \frac{i\,\sigma(\omega)}{\omega}\right)\vec{E}(\omega) = \tilde{\varepsilon}(\omega)\,\vec{E}(\omega)$$

$$\Rightarrow \tilde{\varepsilon}(\omega) = \varepsilon_0\left(1 + \frac{i\,\sigma(\omega)}{\varepsilon_0\,\omega}\right) \xrightarrow[\omega\to 0]{} \frac{i\sigma}{\omega} \quad \text{for conductors } (\sigma \neq 0)$$

$$\longrightarrow \varepsilon_0 \quad \text{for insulators } (\sigma = 0)$$

<u>High frequency behavior:</u>

At very high ω particle's trajectory is very short, $\sim l_T$, and therefore is not sensitive to internal (atomic) fields $\Rightarrow$ particle can be treated as free:

$$m\ddot{\vec{r}} = q\vec{E}_0 e^{-i\omega t} = q\vec{E}(t)$$

$$\Rightarrow \vec{v} = \dot{\vec{r}} = \frac{q\vec{E}_0 e^{-i\omega t}}{-i\omega m}$$

$$\Rightarrow \vec{j}_{md} = nq\vec{v} = i\frac{nq^2}{\omega m}\vec{E}_0 e^{-i\omega t}$$

$$\Rightarrow \vec{\mathcal{P}}(t) = \int_{-\infty}^{t} dt'\,\vec{j}_{md}(t') = -\frac{nq^2}{m\omega^2}\vec{E}_0 e^{-i\omega t}$$

$$\Rightarrow \vec{\mathcal{D}} = \varepsilon_0\vec{E} + \vec{\mathcal{P}} = \varepsilon_0\left(1 - \frac{nq^2}{\varepsilon_0 m\omega^2}\right)\vec{E}_0 e^{-i\omega t}$$

Thus, $\vec{D}(t) = \vec{D}_0 e^{-i\omega t}$ or $\vec{D}_\omega e^{-i\omega t}$

↳ amplitude of field with frequency ω.

$$\vec{D}_\omega = \tilde{\varepsilon}(\omega) \vec{E}_\omega$$

$$\tilde{\varepsilon}(\omega) = \varepsilon_0 \left(1 - \frac{nq^2}{\varepsilon_0 m \omega^2}\right) = \varepsilon_0 \left(1 - \frac{\omega_p^2}{\omega^2}\right)$$

where $\omega_p = \sqrt{\frac{nq^2}{\varepsilon_0 m}}$ is plasma frequency (and for electrons $q = -e$)

Thus we see that $\tilde{\varepsilon}(\omega) \to \varepsilon_0$ for $\omega \to \infty$ but can change sign for $\omega < \omega_p$.

Lorentz – Drude model

$$m\left(\ddot{\vec{r}} + \gamma\dot{\vec{r}} + \omega_0^2\vec{r}\right) = q\vec{E}(r,t) \qquad (q=-e)$$

$\uparrow$ damping force

Assume wave length $\lambda \gg$ typical displacement $\vec{r}$

$$\Rightarrow \vec{E}(r,t) = \vec{E}(\omega)e^{-i\omega t} \Rightarrow \vec{r}(t) = \vec{r}(\omega)e^{-i\omega t}$$

$$\left(-\omega^2 - i\omega\gamma + \omega_0^2\right)\vec{r}(\omega) = \frac{q}{m}\vec{E}(\omega)$$

$$\vec{p}(\omega) = q\vec{r}(\omega) = \frac{q^2/m}{\omega_0^2-\omega^2-i\omega\gamma}\vec{E}(\omega) \Rightarrow \vec{P} = N\vec{p}$$

$$\Rightarrow \varepsilon(\omega) = \varepsilon_0\left(1 + \frac{Ne^2}{\varepsilon_0 m}\sum_j \frac{f_j}{\omega_j^2-\omega^2-i\omega\gamma_j}\right)$$

f_j = number of electrons with ω_j

where N = # number of molecules per unit volume

Z = # electrons per molecule, $Z = \sum_j f_j$

<u>Insulators</u>: the lowest resonant frequency is finite, $\omega_j \neq 0$

$$\Rightarrow \frac{\varepsilon(\omega)}{\varepsilon_0} \xrightarrow[\omega\to 0]{} \left(1 + \frac{Ne^2}{\varepsilon_0 m}\sum_j \frac{f_j}{\omega_j^2}\right)$$

<u>Conductors</u> some fraction, f_0, of electrons are free

$\Rightarrow$ for them $\omega_j = 0$ (no restoring force)

$$\Rightarrow \varepsilon(\omega) = \varepsilon_0\left(1 + \frac{Ne^2}{\varepsilon_0 m}\sum_{j\neq 0}\frac{f_j}{\omega_j^2-\omega^2-i\omega\gamma_j}\right) +$$

$$\varepsilon_0\frac{Ne^2 f_0}{\varepsilon_0 m\omega(\omega+i\gamma)} = \varepsilon_b(\omega) + \frac{i\sigma(\omega)}{\omega}$$

$$\Rightarrow i\sigma(\omega) = -\frac{Ne^2 f_0}{m(\omega+i\gamma)} = -\frac{Ne^2 f_0}{i\gamma m\left(1 - \frac{i\omega}{\gamma}\right)} = i\,\frac{Ne^2 f_0 \tau}{m} \cdot \frac{1}{1-i\omega\tau}$$

$$\gamma = \frac{1}{\tau}$$

Thus obtain $\sigma(\omega) = \dfrac{\sigma_0}{1-i\omega\tau}$ (Drude's result)

where $Nf_0 \equiv n$ = density of free electrons.

Two important limits:

(a) low-frequency (d.c.)

$$\varepsilon(\omega) \to \varepsilon_b(0) + \frac{i\sigma_0}{\omega} \approx \frac{i\sigma_0}{\omega}$$ dominated by $\frac{\sigma_0}{\omega}$ term.

(b) high-frequency

$$\varepsilon(\omega) \to \varepsilon_b(\omega) - \varepsilon_0 \frac{Nf_0 e^2}{\varepsilon_0 m \omega^2} = \varepsilon_b(\omega) - \varepsilon_0 \frac{\omega_0^2}{\omega^2}$$

where $\omega_0^2 = \dfrac{Nf_0 e^2}{\varepsilon_0 m} = \dfrac{ne^2}{\varepsilon_0 m}$ = plasma frequency.

Drude again: Summary

$$\frac{d}{dt}\vec{v}_d = \frac{q}{m}\vec{E}_0 e^{-i\omega t} - \frac{\vec{v}_d}{\tau}$$

$$\vec{v}_d(t) = \vec{v}_d\, e^{-i\omega t}$$

$$-i\omega\,\vec{v}_d + \frac{1}{\tau}\vec{v}_d = \frac{q\vec{E}_0}{m}\cancel{e^{-i\omega t}}$$

$$\vec{v}_d = \frac{q\vec{E}_0}{m\left(-i\omega + \frac{1}{\tau}\right)}$$

$$\vec{v}_d(t) = \frac{q\vec{E}_0 e^{-i\omega t}}{m\left(-i\omega + \frac{1}{\tau}\right)}$$

$$\vec{j}(t) = qn\vec{v}_d = \frac{q^2 n\,\vec{E}_0 e^{-i\omega t}}{m\,\frac{1}{\tau}\left(1 - i\omega\tau\right)} = \sigma(\omega)\,\vec{E}_0 e^{-i\omega t}$$

Drude conductivity $\sigma(\omega) = \dfrac{e^2 n\tau}{m(1-i\omega\tau)} = \dfrac{\sigma_0}{1-i\omega\tau}$

Dielectric function $\varepsilon(\omega) = \varepsilon_0 + \dfrac{i\sigma(\omega)}{\omega}$

$$\varepsilon(\omega) = \varepsilon_0 + \frac{i q^2 n}{m\omega\left(-i\omega + \frac{1}{\tau}\right)} = \varepsilon_0 - \frac{i q^2 n}{i m\omega\left(\omega + \frac{i}{\tau}\right)}$$

$$= \varepsilon_0\left(1 - \frac{q^2 n}{\varepsilon_0 m}\,\frac{1}{\omega\left(\omega + \frac{i}{\tau}\right)}\right)$$

So that $\varepsilon(\omega) = \varepsilon_0\left(1 - \dfrac{\omega_p^2}{\omega\left(\omega + \frac{i}{\tau}\right)}\right)$, $\omega_p = \sqrt{\dfrac{e^2 n}{\varepsilon_0 m}}$

in conductors

plasma frequency

In dielectrics: $\varepsilon(\omega) = \varepsilon_0\left(1 + \dfrac{\omega_p^2}{\omega_0^2 - \omega^2 - i\frac{\omega}{\tau}}\right)$

where ω_0 = restoring frequency ($\omega_0 = 0$ in conductors)

$\varepsilon(\omega)$ and absorption

$$\vec{D}(\omega) = \varepsilon_0 \vec{E}(\omega) + \cancel{\chi(\omega)}\, \vec{P}(\omega) = \varepsilon_0 \left(1+\chi(\omega)\right) \vec{E}(\omega) = \varepsilon(\omega)\, \vec{E}(\omega)$$

$$\hookrightarrow \varepsilon_0 \chi(\omega) \vec{E}(\omega)$$

Dielectric function $\varepsilon(\omega) = \varepsilon_0 \left(1+\chi(\omega)\right)$

Hence $\varepsilon(\omega)$ has the same properties as $\chi(\omega)$:

$$\varepsilon'(\omega) = \varepsilon'(-\omega) \quad , \quad \varepsilon''(\omega) = -\varepsilon''(-\omega).$$

From $\vec{P} = \vec{D} - \varepsilon_0 \vec{E} = (\varepsilon - \varepsilon_0)\, \vec{E}$, $\vec{E}(t) = \vec{E}(\omega) e^{-i\omega t}$

$$\frac{\partial \vec{P}}{\partial t} = \vec{J} = \left(\frac{\partial}{\partial t} \frac{\sum q \vec{r}}{\Delta V} = \frac{\sum q \vec{v}}{\Delta V} \right)$$

$$\Rightarrow -i\omega \vec{P}(\omega) = -i\omega \left(\varepsilon(\omega) - \varepsilon_0\right) \vec{E}(\omega)$$

$$\Rightarrow -i\omega \vec{P}(\omega) = \underbrace{\vec{J}(\omega) = \sigma(\omega)\, \vec{E}(\omega)}_{\text{Ohm's law for conductors}}$$

$$\Rightarrow \vec{P}(\omega) = \left(\varepsilon(\omega) - \varepsilon_0\right) \vec{E}(\omega) = \frac{i\sigma(\omega)}{\omega} \vec{E}(\omega)$$

$$\varepsilon(\omega) = \varepsilon_0 + \frac{i\sigma(\omega)}{\omega} .$$

Connection with energy absorption — Joule heat

power $P = \vec{J} \cdot \vec{E}$

$$\vec{E}(t) = \mathrm{Re}\left\{ \vec{E}(\omega) e^{-i\omega t} \right\} = \frac{1}{2} \left(\vec{E}(\omega) e^{-i\omega t} + \vec{E}^* e^{i\omega t} \right)$$

$$\vec{J}(t) = \mathrm{Re}\left\{ \underset{\sigma(\omega)\vec{E}(\omega)}{\underset{\parallel}{\vec{J}(\omega)}} e^{-i\omega t} \right\} = \frac{1}{2} \left(\sigma(\omega) \vec{E}(\omega) e^{-i\omega t} + \sigma^*(\omega) \vec{E}^*(\omega) e^{i\omega t} \right)$$

$$P = \frac{1}{4}\left(\vec{E}_\omega e^{-i\omega t} + \vec{E}^*_\omega e^{i\omega t}\right)\left(\sigma(\omega)\vec{E}_\omega e^{-i\omega t} + \sigma^*(\omega)\vec{E}^*_\omega e^{i\omega t}\right)$$

$$= \frac{1}{4}\Big(\sigma(\omega)\,\vec{E}_\omega\cdot\vec{E}_\omega e^{-i2\omega t} + \sigma^*(\omega)\vec{E}^*_\omega\cdot\vec{E}^*_\omega e^{i2\omega t}$$

$$+ \left(\sigma(\omega) + \sigma^*(\omega)\right)\vec{E}_\omega\cdot\vec{E}^*_\omega\Big)$$

Average power over the period $\Rightarrow \overline{e^{\pm i2\omega t}} = 0$.

$$\overline{P} = \frac{1}{2}\sigma'(\omega)\,|\vec{E}(\omega)|^2 = \frac{1}{2}\omega\,\varepsilon''(\omega)\,|\vec{E}(\omega)|^2.$$

$$\varepsilon^*(\omega) = \varepsilon_0 + \frac{i}{\omega}\left(\sigma'(\omega) + i\sigma''(\omega)\right)$$

$$\Rightarrow \begin{cases} \varepsilon'(\omega) = \varepsilon_0 - \dfrac{\sigma''(\omega)}{\omega} \\ \varepsilon''(\omega) = \dfrac{\sigma'(\omega)}{\omega} \end{cases}$$

$$\overline{P}(\omega) > 0 \Rightarrow \begin{cases} \varepsilon''(\omega) > 0 & \text{for } \omega > 0 \\ \varepsilon''(\omega) < 0 & \text{for } \omega < 0. \end{cases}$$

Drude model: $\sigma(\omega) = \dfrac{ie^2 n}{m\left(\omega + \frac{i}{\tau}\right)} = \dfrac{e^2 n\tau}{m(1 - i\omega\tau)} = \dfrac{\sigma_0}{1 - i\omega\tau}$

$$\Rightarrow \sigma^*(\omega) = \frac{\sigma_0(1 + i\omega\tau)}{1 + \omega^2\tau^2} \longrightarrow \begin{cases} \sigma_0, & \omega\tau \ll 1 \text{ large absorption} \\ \dfrac{i\sigma_0}{\omega\tau}, & \omega\tau \gg 1 \text{ low absorption} \end{cases}$$